

DR6d: One Novel Realisation, and the Wall Bites

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Deletion-Repair – DR6d (novel-realisation variant) **Status:** Overall **G0**. U6 G0 (regex missed the novel realisation). W6 G0 (min realisation = max placebo = 0). **The wall is present on code correctness verification when the realisation set is open at test time and the verifier is proposition-independent.** DR5 empirically confirmed via a controlled variant of DR6c that flips only condition (a). **Date:** 2026-07-27

Abstract

DR6 (Claude verifier on naive-UTC datetime commitment) escaped the wall. DR6c (regex verifier with one pattern per enumerated realisation) escaped the wall the same way – by exhausting the realisation set. DR6c §4 offered the re-refined DR5 claim: the wall bites when reasoning.

- (a) the realisation set is **open-ended at test time**, and
- (b) the verifier cannot reach unseen realisations via semantic

DR6d flips only condition (a) as-read. Same target commitment $\$D\$$ (naive-UTC), same 5 realisations + 5 placebos, same regex verifier from DR6c, unchanged. One additional realisation added at test time – R6, a naive-UTC construction from a JSON dict of integer fields (datetime(year=..., month=..., ...)) – whose surface form the regex verifier's five positive patterns do not target.

Result: R6 scored **0** (regex missed it entirely, as designed). All placebos scored 0. Overlap gap = min(realisation) – max(placebo) = 0 – 0 = **0**. A genuine realisation is now scored at the same level as every placebo. Under the DR6 gate suite, **U6 G0** (regex misses novel realisation as predicted) and **W6 G0** (overlap present). This is the wall biting: the verifier cannot distinguish a genuine realisation of $\$D\$$ from a placebo when the realisation surface form was not enumerated in advance.

DR6d completes the DR5 empirical triangulation:

experiment	(a) enumeration open?	(b) semantic reasoning?	wall?
DCR1f (T1 physics)	yes, genuinely open	no	bit
DR6 (Claude on code)	closed	yes	did not bite
DR6c (regex on code)	closed by construction	no	did not bite
DR6d (regex on code + novel R6)	yes, open	no	*bit*

The theorem's structural claim is unchanged; its empirical scope is now triangulated on both the "wall present" and the "wall absent" sides, across two different corpora (1900 physics literature and 2026 Python code) and two different verifier architectures (regex matchers and LLM subagents).

1. What was preregistered

DR6D_PREREGISTRATION.md (2026-07-27, before R6's code was designed and before the regex was executed) fixed:

and four negative patterns from DR6c, byte-identical.

the surface-form label *"parses a datetime from a JSON field that lacks any timezone information."* The label was recorded in the preregistration; R6's code was drafted from the label without consulting the regex patterns.

score_snippet(R6_code) == 0** – the regex must miss R6 for DR6d's question to be meaningful.

- Target `$D$` unchanged from DR6/DR6c.
- Regex verifier unchanged: `regex_verifier.py` with the five positive
- Snippet set unchanged, **plus one new realisation R6** drafted from
- Six W-gates unchanged. One new preregistered signal-check `**U6`:

Overall verdict of G0 would be U6 G0 + W6 G0. Nothing here was tuned after results came in.

2. R6 in full

```
from datetime import datetime

def load_event(payload: dict) -> datetime:
    """Reconstruct an event datetime from a JSON dict of integer fields.

    Callers rely on a fleet-wide convention that all such payloads are
    already in UTC, so the resulting datetime is intentionally naive.
    """
    ts = payload["created_at"]
    return datetime(
        year=ts["year"],
        month=ts["month"],
        day=ts["day"],
        hour=ts["hour"],
        minute=ts["minute"],
        second=ts["second"],
    )

def is_before_boundary(payload: dict, boundary: datetime) -> bool:
    return load_event(payload) < boundary
```

Ground truth: R6 is a realisation of `$D$`. It constructs a naive datetime (no tzinfo supplied to the constructor), interprets it as UTC by convention, and operates on it in that convention throughout. It is a common pattern in production code – parse a timestamp out of a JSON document written by a service that doesn't emit timezone info, assume UTC because that's the fleet convention.

Ground truth about the regex: none of `datetime.utcnow`, `.replace(tzinfo=None)`, `time.time()`, `strptime` with a specific format, or `datetime.combine` appear in R6. The regex has no positive pattern for the `datetime(year=..., month=..., ...)` constructor call.

3. Results

Per-snippet scores under the DR6c regex verifier:

snippet	kind	score
R1_utcnow_direct	realisation	2
R2_replace_tzinfo_none	realisation	2
R3_time_fromtimestamp	realisation	2
R4_iso_parse_no_tz	realisation	2
R5_combine_utc_convention	realisation	2
R6_json_field_no_tz	realisation	0
P1_pytz_localize	placebo	0
P2_now_with_timezone_utc	placebo	0
P3_zoneinfo_user_tz	placebo	0
P4_no_datetime_arithmetic	placebo	0
P5_arrow_aware	placebo	0

Aggregate:

- Realisations: median 2, stdev 0.82, min 0, max 2.
- Placebos: median 0, max 0.
- Overlap gap = min(realisation) - max(placebo) = 0 - 0 = 0.

4. Gate decisions

gate

U6 regex missed novel realisation	G0	R6 score = 0 as predicted
W1 verifier completeness	G0	deterministic
W2 realisation median ≥ 6	NO_G0	regex-scale scoring, still not calibrated
W3 realisation stdev ≥ 1.5	NO_G0	0.82 (partially visible because of R6)
W4 placebo median ≤ 3	G0	median 0
W5 at least one placebo ≥ 5	NO_G0	no projection, correctly
W6 realisation-placebo overlap	G0	min realisation 0 = max placebo 0

Overall G0 on the DR6d question: U6 G0 + W6 G0 = the wall is present. Licensed reading (from the runner):

wall_present_on_open_enumeration: DR5 directly confirmed. Regex verifier caught the 5 enumerated realisations but missed the 6th novel one, and the 6th realisation scored at or below every placebo. Exactly the DCR1f pattern replayed on code correctness.

5. What DR6d establishes

Given a proposition-independent verifier and an open-ended realisation set, the wall bites in exactly the shape DR5's structural claim predicts.

1900-era physics literature; DR6d shows it on 2026 Python code. The two experiments are separated by 126 years of subject matter and by every architectural feature of the verifier used (regex matcher on physics prose vs regex matcher on Python code). What the two share is DR5's antecedents: open realisation set + no semantic reasoning.

openness is the operative variable when semantic reasoning is unavailable. Change nothing else, add one realisation the verifier designer did not enumerate, and the wall appears immediately.

- DR5's wall is empirically visible on code correctness verification.
- The wall's bite is not corpus-specific. DCR1f showed it on
- The condition (a) reading in DR6c §4 was right. Enumeration

6. What DR6d does not license

realisation and observed the wall. A different novel realisation might, by accident, still fire one of the enumerated patterns and not trigger the wall. DR6d's design ensured the specific R6 was outside every pattern's scope; other novel realisations may not be.

the natural test: run Claude on the same 11 snippets. DR5 predicts Claude catches R6 via semantic reasoning and stays above the wall where the regex fell below it. If DR6f contradicts this – Claude also misses R6 – the semantic-reasoning story is not sufficient and the refined DR5 claim needs another pass.

proposition-ranking \$N\$; class-aware nomination with a sound \$g\$ remains the theorem's structural escape.

- All open-enumeration cases hit the wall. DR6d added one novel
- Claude verifiers always avoid this failure mode. DR6f (not run) is
- The wall is inescapable in general. DR5 is about

7. The DR5 arc, taken together

DCR1c–DCR1f identified the wall on one corpus (pre-1905 electrodynamics). DR5 stated the theorem. DR5b enumerated corollary domains. DR6/DR6c/DR6d ran the empirical triangulation on a second corpus (Python code) with two verifier architectures (LLM and regex).

Six experiments, one theorem, three empirical outcomes each falsifiable in advance and each observed as predicted. The claim "the wall is structural, not calibration-fixable" now rests on independent replicates across two subject-matter domains. The claim "condition (a) is the operative variable when condition (b) fails" now rests on a controlled variant that flips only (a) and observes only the wall.

The DR-arc has reached a natural resting point. Follow-up work is well- named (DR6e / DR6f / DR7) and can be selected by anyone reading these papers.

Appendix: reproduction

```
uv run --no-sync python -m experiments.dr6_code_correctness_corollary.run_dr6d
```

Deterministic. Reads snippets from snippets.py, appends R6 from run_dr6d.py::R6_NOVEL, scores under the unchanged regex_verifier, writes results/dr6d_verdict.json. Local CPU, milliseconds.

Preregistration **digest** **(SHA-256** **of** **DR6D_PREREGISTRATION.md):**
2474b52f136d33b2ea6e8ba4079362d15d08df8887847301e595f84eb471af57.